

## USTI NAD ORLICI, Czech Republic

WMO: 116790

Lat: 49.9803N

Lon: 16.4222E

Elev: 405

StdP: 96.55

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -13.9      | -11.1 | -17.2                         | 0.9 | -12.9 | -14.3 | 1.1 | -10.3 | 13.5                  | 3.4  | 11.9 | 3.0  | 1.8                   | 70   | 0.569 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.0                   | 30.0            | 19.5 | 28.1 | 18.7 | 26.3 | 17.9 | 20.5                | 27.5 | 19.6 | 26.2 | 18.7 | 24.7 | 3.3                  | 120  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 18.2                            | 13.7 | 22.9 | 17.3 | 13.0 | 21.8 | 16.6 | 12.4 | 21.0 | 60.5          | 27.5 | 57.6 | 26.4 | 54.8 | 24.9 | 26.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
| (a)               | (b)  | (c) | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| 10.2              | 8.8  | 7.7 | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
|                   |      |     | -17.7                      | 32.9 | 4.1                | 1.8 | -20.6                                              | 34.2 | -23.0      | 35.3 | -25.3      | 36.3 | -28.3      | 37.6 |
|                   |      |     | -18.0                      | 22.6 | 4.0                | 1.5 | -20.9                                              | 23.7 | -23.3      | 24.5 | -25.5      | 25.3 | -28.5      | 26.4 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 8.5    | -1.9  | -0.4  | 3.3   | 8.6   | 13.2  | 16.5  | 18.4  | 18.2  | 13.5  | 8.7   | 3.9   | -0.6 |
|                                                                                  | DBStd       | 8.25   | 4.81  | 4.73  | 3.97  | 4.34  | 3.73  | 3.61  | 3.32  | 3.32  | 3.48  | 4.00  | 4.15  | 4.43 |
|                                                                                  | HDD10.0     | 1556   | 370   | 292   | 210   | 75    | 12    | 1     | 0     | 0     | 8     | 73    | 185   | 329  |
|                                                                                  | HDD18.3     | 3707   | 628   | 525   | 467   | 292   | 164   | 78    | 41    | 43    | 150   | 299   | 432   | 588  |
|                                                                                  | CDD10.0     | 1006   | 0     | 0     | 1     | 34    | 111   | 195   | 262   | 254   | 112   | 33    | 3     | 0    |
|                                                                                  | CDD18.3     | 114    | 0     | 0     | 0     | 1     | 4     | 22    | 44    | 39    | 4     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 1233   | 0     | 0     | 0     | 8     | 60    | 241   | 463   | 416   | 46    | 0     | 0     | 0    |
|                                                                                  | CDH26.7     | 333    | 0     | 0     | 0     | 1     | 7     | 58    | 141   | 121   | 6     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 3.2    | 3.8   | 3.6   | 3.7   | 3.3   | 3.1   | 2.8   | 2.8   | 2.5   | 2.7   | 3.2   | 3.6   | 3.7  |
| Precipitation                                                                    | PrecAvg     | 708    | 49    | 41    | 51    | 41    | 71    | 76    | 101   | 75    | 62    | 44    | 48    | 50   |
|                                                                                  | PrecMax     | 885    | 88    | 83    | 138   | 93    | 146   | 138   | 262   | 197   | 140   | 110   | 96    | 90   |
|                                                                                  | PrecMin     | 565    | 18    | 11    | 11    | 6     | 17    | 16    | 25    | 32    | 17    | 6     | 0     | 20   |
|                                                                                  | PrecStd     | 91     | 19    | 22    | 27    | 20    | 30    | 26    | 52    | 40    | 36    | 23    | 22    | 20   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 10.3  | 11.3  | 17.2  | 24.5  | 27.5  | 31.0  | 32.7  | 33.0  | 27.2  | 21.0  | 14.8  | 9.8  |
|                                                                                  |             | MCWB   | 7.5   | 7.5   | 10.2  | 14.5  | 17.3  | 20.4  | 19.6  | 20.3  | 17.5  | 15.3  | 11.5  | 8.1  |
|                                                                                  | 2%          | DB     | 7.8   | 9.0   | 14.4  | 21.1  | 25.0  | 28.5  | 30.1  | 29.7  | 24.5  | 18.3  | 12.8  | 8.0  |
|                                                                                  |             | MCWB   | 5.9   | 6.6   | 8.6   | 12.7  | 16.3  | 19.2  | 19.5  | 19.2  | 16.8  | 13.8  | 10.6  | 6.5  |
|                                                                                  | 5%          | DB     | 5.7   | 7.3   | 11.8  | 18.8  | 22.8  | 26.1  | 28.0  | 27.6  | 22.3  | 16.5  | 11.3  | 6.4  |
|                                                                                  |             | MCWB   | 4.3   | 5.3   | 7.4   | 11.7  | 15.3  | 18.0  | 18.7  | 18.8  | 15.8  | 13.2  | 9.3   | 5.2  |
|                                                                                  | 10%         | DB     | 4.1   | 5.6   | 9.5   | 16.6  | 20.7  | 24.0  | 26.0  | 25.5  | 20.0  | 14.7  | 9.8   | 4.9  |
|                                                                                  |             | MCWB   | 3.0   | 4.0   | 6.4   | 10.6  | 14.2  | 17.0  | 18.1  | 17.7  | 14.7  | 12.1  | 8.2   | 3.9  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 7.9   | 8.2   | 11.0  | 15.1  | 18.8  | 21.6  | 21.7  | 21.4  | 18.9  | 16.0  | 12.1  | 8.2  |
|                                                                                  |             | MCDB   | 9.8   | 10.3  | 15.4  | 22.4  | 25.6  | 28.9  | 28.5  | 28.9  | 24.6  | 19.6  | 13.8  | 9.5  |
|                                                                                  | 2%          | WB     | 6.1   | 6.9   | 9.4   | 13.6  | 17.2  | 20.1  | 20.4  | 20.2  | 17.5  | 14.5  | 10.8  | 6.7  |
|                                                                                  |             | MCDB   | 7.4   | 8.8   | 13.3  | 19.9  | 23.2  | 26.6  | 27.2  | 27.6  | 22.7  | 17.4  | 12.6  | 7.9  |
|                                                                                  | 5%          | WB     | 4.5   | 5.5   | 8.0   | 12.2  | 16.0  | 18.9  | 19.6  | 19.3  | 16.4  | 13.4  | 9.5   | 5.2  |
|                                                                                  |             | MCDB   | 5.7   | 7.1   | 10.9  | 17.6  | 21.6  | 24.8  | 26.2  | 25.9  | 20.8  | 16.0  | 11.0  | 6.2  |
|                                                                                  | 10%         | WB     | 3.1   | 4.1   | 6.7   | 11.0  | 14.8  | 17.7  | 18.7  | 18.5  | 15.4  | 12.4  | 8.3   | 4.0  |
|                                                                                  |             | MCDB   | 4.1   | 5.5   | 9.2   | 15.8  | 19.6  | 22.9  | 24.7  | 24.1  | 19.2  | 14.5  | 9.7   | 4.8  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 4.7   | 6.2   | 7.9   | 10.3  | 10.7  | 10.8  | 11.0  | 11.6  | 9.6   | 7.4   | 4.9   | 4.2  |
|                                                                                  |             | MCDBR  | 5.8   | 8.0   | 12.9  | 14.3  | 14.4  | 14.5  | 15.3  | 15.1  | 13.5  | 10.2  | 6.8   | 5.3  |
|                                                                                  |             | MCWBR  | 4.7   | 5.8   | 7.8   | 7.3   | 7.1   | 6.6   | 6.1   | 6.1   | 6.5   | 6.3   | 4.9   | 4.4  |
|                                                                                  | 5% WB       | MCDBR  | 5.7   | 7.2   | 11.0  | 12.8  | 13.0  | 13.4  | 13.2  | 13.2  | 11.8  | 9.3   | 6.1   | 5.1  |
|                                                                                  |             | MCWBR  | 4.8   | 5.6   | 7.3   | 6.8   | 6.8   | 6.5   | 5.8   | 5.9   | 6.2   | 5.9   | 4.6   | 4.4  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.302  | 0.326 | 0.367 | 0.413 | 0.413 | 0.412 | 0.427 | 0.421 | 0.388 | 0.361 | 0.335 | 0.300 |      |
|                                                                                  | taud        | 2.324  | 2.298 | 2.251 | 2.182 | 2.225 | 2.259 | 2.223 | 2.254 | 2.317 | 2.360 | 2.372 | 2.370 |      |
|                                                                                  | Ebn at Noon | 733    | 805   | 830   | 828   | 845   | 848   | 826   | 812   | 801   | 754   | 685   | 682   |      |
|                                                                                  | Edh at Noon | 68     | 89    | 112   | 134   | 134   | 130   | 133   | 123   | 104   | 84    | 66    | 57    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.84   | 1.61  | 2.68  | 4.17  | 5.02  | 5.35  | 5.09  | 4.57  | 3.23  | 1.88  | 0.92  | 0.64  |      |
|                                                                                  | RadStd      | 0.15   | 0.23  | 0.44  | 0.52  | 0.60  | 0.49  | 0.56  | 0.42  | 0.47  | 0.28  | 0.11  | 0.07  |      |

## Historical Trends

|                         |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                         |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Station Only            |  | +0.43 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -129    | +51     | +19     |
| Regional (55 neighbors) |  | +0.49 | N/A     | N/A    | N/A     | +0.35 | +0.32 | N/A         | -153    | +65     | +25     |

Nomenclature: See separate page